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Physical Research Laboratory Ahmedabad

SUMMER INTERNSHIP PROJECT REPORT

CERTIFICATE

This is to certify that the work embodies in this summer internship project entitled “**Proto-Planetary Disc Evolution and Early Stellar Growth**” being submitted by “**Ali Mur-taza**” of **Indian Institute Of Technology, Bombay** for partial fulfillment of the requirement for the award of “Certificate” during the period of 2th June to 30th July PRL’s Summer internship programme 2026 is a record of a bonafide and original piece of work, which was carried out by him under my supervision and guidance in the Astronomy and Astrophysics Division of PRL, Ahmedabad.

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Date:
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Abstract

This report details the implementation of automated spectral line profile fitting pipelines to determine the instantaneous mass accretion rates ($\dot{M}_{\text{acc}}$) of Young Stellar Objects (YSOs). Utilizing high-resolution optical and near-infrared spectroscopy, we isolate key diagnostic emission features, correct for interstellar dust extinction using parameter-average laws, and map line luminosities to total accretion energy. Here, we evaluate the performance, reliability, and systematic variations of automated line-fitting architectures against verified empirical samples to establish a streamlined, reproducible workflow for young stellar population analysis.

1 Introduction

The evolution of pre-main-sequence stars is fundamentally governed by the interplay between material accretion from surrounding circumstellar disks and powerful magnetospheric outflows. Accurately quantifying the mass accretion rate ($\dot{M}_{\text{acc}}$) provides vital constraints on disk dissipation timescales and planet formation timelines. This project focuses on isolating absolute line fluxes from optical/NIR spectra to track these energetic processes.

2 Literature Review

[Facchini *et al.* 2016](#) study RW Aur, which is a binary system undergoing a tidal interaction between its primary and circumstellar disc. It has gone through two consecutive dimming events ($\Delta\text{mag} \approx 2$ in V band). They study the spectra of this system during the dimming epoch using low-resolution X-shooter data and previously available high-resolution UVES data. They particularly use two lines Na and K and found out in the dimming phase there are two narrow absorption lines in Na - K doublet with a velocity of -60 Km/s where the Na is saturated. A line being saturated means that the column density is so thick that all of the light is being absorbed at that particular wavelength. This indicates that during the dimming phase, the disk wind is either significantly enhanced or overlaps directly with line of sight. The photospheric evolution during dimming shows a grey extinction which is correlated with reduction of Equivalent Width (EW) across all wavelengths, and the tracers for accretion do not vary across the dimming event. They concluded that this dimming phase could be caused by a perturbation of the inner disk. They also suggest that the inner disk is occulting most of the star and thus its photosphere but not the accretion disk within a few stellar radii. Since, the outerdisk is modestly inclined, they proposed an unseen sub-stellar companion, which would also explain the 2.77 periodic variability in the spectral lines.

[Cahill *et al.* 2019](#) have studied T Chamaeleontis (T Cha) which is a transitional disk (TD) object using spectro-astrometry to understand whether it can be used to confirm the existence of a proto-planet T Cha b. They used direct accretion tracers like H α and indirect tracers like [OI] 6300 Angstrom line during four epochs from UVES data. [OI] is classified as a broad low velocity component (BC LVC). They found that wind was likely the origin of [OI] and also that [OI] was a better accretion tracer than H α in this case. This shows

the T cha may not be a weak accretor but when comparing the $\dot{M}_{\text{acc}}$ calculated from $L[\text{OI}]$ and $L_{\text{H}\alpha}$ shows that a significant proportion of the $\text{H}\alpha$ emission is suppressed during periods of obscuration. This verifies that the bulk origin of $\text{H}\alpha$ emissions are accretion shocks. Furthermore, they also say that spectro-astrometric measurements, using the $\dot{M}_{\text{acc}}$ for $\text{H}\alpha$ for T cha b are not sufficient to rule out the existence of a companion T cha b, as its accuracy is not sufficient to confirm a detection. As the $\text{H}\alpha$ emission and the continuum emission are both variable, it is very difficult to apply SA to this source. As the line is often in absorption, it is often very hard to quantify it and subtract it from the continuum.

(Armeni *et al.*, 2023) investigate the inner disk structure and accretion variability of star HM Lup, a young classical T Tauri Star. They used a multi-wavelength approach using different instruments across the optical and UV spectra simultaneously using HST/STIS, VLT/Xshooter, alongside photometric data from TESS and AAVSO, by fitting the UV continuum excess and integrating various emission lines to measure mass accretion rate. They discovered that HM Lup is a strong accretor and it transitioned from a chaotic accretor (2019) to an unstable ordered regime in 2021. The 2021 high-accretion rate was characterized by "accretion buster" quasi-periodic variability in its light-curve. Their spectroscopic analysis reveals that broad hydrogen (H I) and Helium (He I) lines originate in hot ($\approx 10,000$ K) inner pre-shock region close to the star, whereas the metallic lines such as Fe I, Fe II, Ca II, Na I form in a cooler (≈ 6000 K) region near the disk. They conclude that the outbursts were driven by an increased mass accretion rates that compressed the magnetosphere and caused density enhancement in the outer accretion flow, which in turn produced enhanced emission seen in the metallic species during the outburst.

Fiorellino & Somigiana, 2026 have provided a detailed review on the process of accretion onto protostars from accretion disk, from the evolution of the envelope to formation of disk and the ensuing processes that happen, like Boundary layer (BL) accretion and magnetospheric accretion (MA), both differing depending on the age and class of the envelope. They have shown that most of the current studies surrounding accretion onto protostars involves assumptions made for Class II objects, for which accretion is greatly understood, however application of these studies, which involve using empirical relations between line luminosity (luminosity of a particular emission line from the source (L_{line})) and the accretion luminosity (L_{acc} for Class O and I sources), remains poorly understood, and applying these relations naively to them will give inaccurate results. As constraining the mass of a disk in earlier stages is difficult due to optical depth effects. The X-wind model which couples the magnetospheric accretion to the jet launching and the inner disk truncation radius, through this process the accretion process not only constrains the mass transfer but also the angular momentum budget, thermal and dynamical evolution of the inner disk.

We will focus on summarizing each section from this paper, as this offers the most comprehensive information of the process of accretion onto protostars:

2.0.1 Classification of YSO

Collapse of a prestellar core into a star onsets when the core mass exceeds the Bonnor-Ebert critical mass, causing gravitational collapse. The first collapse -> increasing density and temperature creates the hydrostatic core, which is a transient, pressure supported object which temporarily halts further collapse. The onset of the second collapse then leads to

the formation of protostar surrounded by thick circumstellar disk, marking the beginning of protostellar phase, during which the forming star stays embedded in its optically thick envelope, and accretes a substantial fraction of its final mass. By the time the envelope dissipates, most of the stellar mass has been assembled, ending the protostellar phase. Young stellar objects (YSOs) are classified into categories based on their infrared spectral index, computed between 2 to 24 μm as $\alpha_{\text{IR}} = \frac{d \log(\lambda F_{\lambda})}{d \log(\lambda)}$ (which basically tells the excess radiation in IR, that the system is producing), bolometric temperature, T_{bol} (which is the temperature of a black body producing radiation at the same mean frequency) the ratio of submillimeter to bolometric luminosity, $L_{\text{submm}}/L_{\text{bol}}$, where L_{submm} is defined as the integrated luminosity at wavelengths $\lambda \geq 350 \mu\text{m}$.

The YSO are categorized based on their observational signatures and evolutionary stages of the accreting young stars as follows:

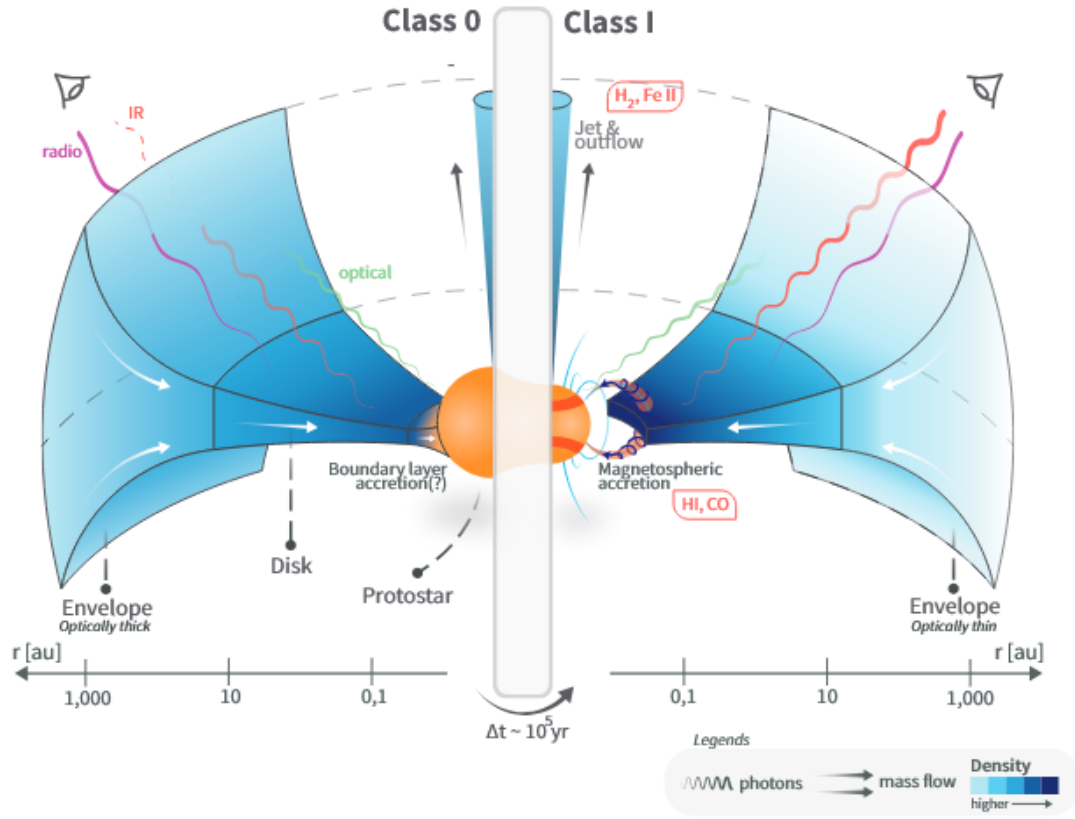


Figure 1. Schematic representation of the accretion process in Class 0 and Class I protostars. Class 0 objects (left) are characterized by a massive, optically thick envelope dominating the system mass and feeding the disk and protostar, with accretion possibly occurring through non-magnetospheric inner-disk processes. In Class I sources (right), the envelope becomes progressively thinner and less massive, the disk plays a more central role, the magnetic field get stronger, and magnetospheric accretion may emerge, producing accretion diagnostics such as H α and CO emission. The figure summarizes the evolution of density structure, mass-flow geometry, and observational tracers across the protostellar phase. In both stages, the accretion process is expected to be accompanied by powerful jets and outflows, traced by shock-excited species such as H $_2$ and [FeII], which play a key role in extracting angular momentum from the system. Figure credits: Dr. Adrien Houge.

Figure 1: Schematic representation of the accretion process in Class 0 and Class I protostars. Class 0 objects (left) are characterized by a massive, optically thick envelope dominating the system mass and feeding the disk and protostar, with accretion possibly occurring through non-magnetospheric inner-disk processes. In Class I sources (right), the envelope becomes progressively thinner and less massive, the disk plays a more central role, the magnetic field get stronger, and magnetospheric accretion may emerge, producing accretion diagnostics such as H α and CO emission. The figure summarizes the evolution of density structure, mass-flow geometry, and observational tracers across the protostellar phase. In both stages, the accretion process is expected to be accompanied by powerful jets and outflows, traced by shock-excited species such as H $_2$ and [FeII], which play a key role in extracting angular momentum from the system. Figure credits: Dr. Adrien Houge. [Fiorellino & Somigliana \(2026\)](#)

- Class 0 (Early protostellar phase): These are the earliest stage of a protostellar phase, the envelope is spherical and optically thick, the magnetic field of the star may not be mature enough to cause MA, there a possible mechanism of accretion is BL accretion. They are characterized by $\alpha \geq 0.3$ [Fiorellino & Somigliana \(2026\)](#) and $T_{\text{bol}} < 70K$. Statistical studies suggest Class 0s have a life time of $\sim 10^4 - 10^5 \text{yr}$. The mass accretion rate in these is theorized to be higher than Class II, episodic bursts of accretion are thought to be intense and frequent.
- Class I (Later Protostellar Phase): Class I sources are protostars still embedded in envelopes but are optically thin than around class 0 objects. They have $\alpha \geq 0.3$ and $70 < T_{\text{bol}} < 650K$; whether the bulk of stellar mass is already assembled at this stage is uncertain. The lifetime of class I is expected to last a few 10^5yr . Like class 0, they are expected to have very high accretion rates as well as frequent and intense episodic bursts.
- Flat-spectrum (FS) (Transition between phases): FS sources are generally interpreted as transition between protostellar and Pre-main sequence (PMS) phase. They have $-0.3 < \alpha < 0.3$ and $650 < T_{\text{bol}} \leq 2800K$, they have likely assembled their final stellar mass, their lifetimes are $\sim 10^5 \text{yr}$ and the mass accretion rate is relatively low with episodic bursts being rare.
- Class II (PMS phase); Class II objects, also known as Classical T Tauri Stars (CTTSs), are PMS stars surrounded by circumstellar disks but largely devoid of envelope. Their SEDs are characterized by $\alpha \leq -0.3$ and $T_{\text{bol}} > 2500K$, their stellar mass are essentially set (although they are still accreting material from disk). The class II phase typically lasts $\sim 2\text{-}3 \text{ Myr}$.
- Class III (PMS Phase): Class III represent latest stage of star formation process, They have dispersed most of their circumstellar disk and show little to no infrared excess, corresponding to weakly accreting or non-accreting young stars, this stage extends several Myr, with typical timescales of $\gtrsim 5\text{-}10 \text{ Myr}$.

The straightforward interpretation of observational classes in terms of evolutionary stages requires caution, the classification is based on SED diagnostic features and from that we infer the lifetime of the phase, this can cause error in lifetime estimates, as it is directly linked to how we classify the phases based on the SED. Timescales for proto and PMS phases are critical for understanding how stellar and planetary mass is assembled. To date most widely adopted lifetimes are from IR photometric classification primarily from *Spitzer* surveys. This while being robust, intrinsically assumes that photometric class assignment is accurate to track evolutionary stages. Near-IR spectroscopic followups of *Spitzer*/2MASS-selected samples have suggested a non-negligible samples are misidentified as protostars but they show spectroscopic signatures of more evolved CTTS. An example is **HOPS 315**, identified as class I based on its bolometric temperature ($T_{\text{bol}} \sim 180K$) but showing a molecular jet typical of younger objects, suggesting it might be a class 0 protostar. A complementary perspective is to use the ratio of compact (disk) and extended (envelope) $870 \mu\text{m}$ emission [Fiorellino & Somigliana \(2026\)](#)—measured by the Atacama Compact Array (ACA) and the

Atacama Large Millimeter Array (ALMA)—which can serve as an evolutionary indicator that is less sensitive to inclination and extinction effects than traditional SED-based classification. It does not redefine the Class 0/I/FS categories but rather provides an independent structural tracer that statistically correlates with them. In their sample, Class 0 protostars are predominantly envelope-dominated (with a disk-to-envelope flux ratio $R < 0.5$), FS sources are largely disk-dominated ($R > 0.5$), and Class I objects span both regimes ($R \in (0, 1)$), suggesting that structural measures of the disk–envelope configuration can robustly trace the evolutionary progression. progression [Fiorellino & Somigliana \(2026\)](#). In the galaxy NGC 1333 sample, 7 out of 17 objects that were identified as class I based on their SEDs were found to be more consistent with class II sources when spectroscopic diagnostics were considered. In the simplified interpretation, objects younger than $\sim 10^4$ yr are usually classified as class 0; Class I/FS correspond to ages between 10^4 and 10^6 yr; finally, class II refer to ages older than 10^6 yr. Stage has been introduced to denote classification based on intrinsic properties- such as relative mass of envelope or disk or envelope accretion rate- rather than on spectral index. Stage 0/I sources are defined by presence of substantial infalling envelope (e.g., $M_{\text{env}} \gtrsim M_{\star}$ or $M_{\text{env}}/M_{\star}$ above a given threshold), independently of their observed SED shape. Mapping observational classes onto physical stages remains non-trivial. Conflating observational class with evolutionary time can be problematic, for example, a $1 M_{\odot}$ star is significantly more evolved than a $0.1 M_{\odot}$ star, since PMS contraction is strongly mass-dependent. For comparison with simulations, it is also non-trivial, an alternative way is based on the envelope mass fraction $M_{\text{env},f}$, defined as the ratio of the envelope mass over total (envelope + sink) mass; in this framework, Class 0 corresponds to $M_{\text{env},f} > 0.5$, while Class I is $0.1 < M_{\text{env},f} < 0.5$.

2.0.2 Observations

Because protostars are deeply embedded in their natal envelopes, optical observations are severely limited. IR and radio wavelengths, where the radiation from forming star is reprocessed and re-emitted by dust in the disk and envelope, are better suited for observing these objects; however, not all IR (especially NIR) emission arises from purely reprocessed radiation. At wavelengths around $\sim 2\mu\text{m}$ part of the observed flux may consist of direct or scattered photospheric emission, as well as hydrogen recombination lines tracing accretion, as a consequence of the local reprocessing of UV/optical photons or their observation in scattered light through outflow cavities. The accretion luminosity (L_{acc}) quantifies the energy released as the material accretes from the circumstellar disk onto the forming star. For CTTS, L_{acc} has been constrained for large samples by modeling the UV continuum excess emission generated by accretion shocks above the stellar photosphere, as well as empirical correlations between L_{acc} and U-band excess luminosity. However, using this strategy for embedded protostars is challenging due to high visual extinction A_v associated with dense envelope, strong continuum veiling (τ) requires detection of stellar photosphere, which is impossible due to the luminosity photosphere overshadowed by accretion luminosity. Alternatively, the accretion luminosity can be derived from emission lines luminosity (L_{line}), such as H I, He I, Fe I, Ca I, Na I, and O I, using empirical relations between L_{acc} and L_{line} [Fiorellino & Somigliana \(2026\)](#). It is important to stress, however, that a tight $L_{\text{acc}}-L_{\text{line}}$ correlation does not necessarily imply a direct physical or causal link between the line-emitting region and the accretion shock

itself. Moreover, interferometric studies have shown that hydrogen recombination lines such as Br γ and H α do not always originate exclusively in magnetospheric funnel flows, but can also trace disk winds or more extended inner-disk regions, while still preserving the empirical $L_{\text{acc}}-L_{\text{line}}$ calibration. Similarly, there has been evidence of empirical relations between L_{acc} and the luminosity of forbidden lines such as [O I] λ 6300, whose origin is clearly associated with outflows rather than direct accretion. These examples highlight that an empirical calibration should not be automatically interpreted as evidence for a specific accretion geometry or emission mechanism.

In most cases dust in the protostellar disk and the envelope absorbs the stellar and accretion luminosity at optical wavelengths and re-emits at longer wavelengths. At NIR band, in particular, emit a number of commonly used tracers which provide a powerful tool to investigate both stellar and accretion properties of protostars. The main ones are (i) strong H I lines tracing accretion, such as Pa β (1.282 μm) and Br γ (2.166 μm), formed in the hot ($T \sim 10^4$ K), dense gas of accretion columns; (ii) CO band-heads in the K -band, which probe the warm inner disk region; (iii) wind and jet tracers, such as [Fe II] at 1.64 μm and H $_2$ lines at 2.12 and 2.24 μm , which are also indirectly linked to accretion; (iv) stellar photospheric features, which allow spectral-typing and stellar parameters constraining; and finally (v) the J -band magnitude, which can be used to estimate the stellar luminosity ($L_\star$). Under the assumption that accretion onto protostars proceeds via magnetospheric accretion, as in CTTs, stellar parameters such as the stellar radius ($R_\star$) and mass ($M_\star$) are required not only to gain a deeper understanding of the protostellar phase, but also to compute the mass accretion rate:

$$\dot{M}_{\text{acc}} \sim \left(1 - \frac{R_\star}{R_{\text{in}}}\right)^{-1} \frac{L_{\text{acc}} R_\star}{GM_\star}, \quad (1)$$

where G is the gravitational constant and R_{in} is the inner radius, usually assumed to be $\sim 5 R_\odot$ for CTTs [Fiorellino & Somigliana \(2026\)](#). Although these observations trace the spatial extent of Br γ emission rather than magnetosphere directly, the inferred sizes provide an observational estimate of inner disk radius R_{in} , commonly associated with the magnetospheric truncation radius in the magnetospheric accretion framework. In addition, recent spectroscopic modeling suggests that the inner disk radius in CTTs may lie as close as $\sim 2-3 R_\star$, smaller than the canonical $\sim 5 R_\star$ often assumed in magnetospheric accretion prescriptions.

2.0.3 Stellar Parameters

We refer to stellar parameters as the quantities that describe the intrinsic properties of protostar, such as effective temperature (T_{eff}), surface gravity ($\log g$), stellar luminosity ($L_\star$), radius ($R_\star$) and mass ($M_\star$). We also include visual extinction (A_V measured along line of sight of star), it is not an intrinsic property of the protostar, but is required to correctly estimate the parameters.

To determine the T_{eff} , A_V and $\log g$ we find spectral type of the protostar by comparing the spectral emission features with a more evolved, like type III, template spectra of a star. We find the $M_\star$ and $R_\star$ by putting the $L_\star$ from the optical and J band in the Hertzsprung-Russell diagram. This is the most widely used method of determining the stellar parameters, this works best for systems where we can see resolve the spectra well and high SNR are available.

Another method to determine protostellar mass is using interferometric observations, reaching angular resolutions of $\sim 0.05''$, corresponding to ~ 10 AU in nearby star forming regions. The stellar mass of protostars can be estimated from ALMA observations by modeling the rotation of molecular gas under keplarian motion assumption, whereby the gas dynamics is dominated by gravitational potential of the central object, where the azimuthal velocity of the gas is given by

$$v_{\phi}(r) = \sqrt{\frac{GM_{\star}}{d_r}},$$

where d_r is the radial distance from the central source. This works well for class II systems as the circumstellar disk is well defined and isolated, for class I and more embedded sources, as there are multiple components, like disk rotation, envelope infall and projection effects, which can cause deviations from purely keplarian motion. Reliable estimate of mass in class I systems require high angular resolution. For Class 0 protostars, the situation is even worse, as the keplarian profile of the disk is largely absent or confined to very small radii and is dominated by a massive envelope. In these systems the gas kinematics are often governed by angular momentum conservation and infall. Consequently, the stellar masses observed from ALMA for class 0 sources are often reported as upper or lower limits or total dynamical masses rather than direct measurement of mass of the protostar alone. We can also fit the spectral energy density (SED) to constrain the stellar parameters, it applies well sampled SED from multi-wavelength observations to a theoretical model. In practice, SED fitting primarily constrains the luminosity attributed to the heating of the disk and envelope, separating L_{acc} from $L_{\star}$ however is model dependent and often degenerate. It is used when high quality observations in the required wavelength are not available and is particularly common for class 0 sources, for which direct constraints on stellar parameters from spectroscopy are often unavailable.

2.0.4 Veiling

Veiling refers to the phenomenon where spectroscopic absorption lines to be weaker than expected in the stellar spectrum. This is due to the extra continuum emission superimposed on the photospheric spectrum. The veiling at a given wavelength λ is defined as

$$r_{\lambda} = \frac{F_{\text{excess}}}{F_{\text{photosphere}}}, \quad (2)$$

where F_{excess} is the excess continuum flux and $F_{\text{photosphere}}$ is the intrinsic photospheric flux at wavelength λ .

In accreting systems, the excess continuum responsible for veiling is produced by the release of accretion energy. In accordance with the magnetospheric accretion models, the inner disk is truncated by the stellar magnetic field at a distance of several stellar radii. Material is then channeled along magnetic field lines, free-falling toward the stellar surface where it terminates in an accretion shock. These accretion hotspots produce largely featureless continuum excess and fuels intense emission lines such as $\text{H}\alpha$, $\text{H}\beta$, and the Ca II Infrared Triplet (IRT). In Boundary layer accretion (BL) the excess continuum emission arises from

the braking and viscous dissipation of keplarian disk material at disk-stellar interface, rather than from localized accretion shocks.

2.0.5 The CO bands

Accreting YSO show CO in both emission and absorption. CO Absorption arises when there is cool gas present between line of sight to a hotter background continuum source. In protostars, this absorpction can occur at different locations, like the stellar photosphere (late-type stars), the cool disk atmosphere (the outer disk or disk surface), and inner disk envelope in class I sources. The physical condition for CO absorption includes The physical conditions for CO absorption include:

1. Temperatures of $\sim 2000\text{--}4000\text{ K}$, which are typically needed to significantly populate the vibrational levels responsible for strong band-head emission.
2. A high CO column density, and
3. A background continuum dominated by the stellar photosphere, hot dust emission, or the accretion continuum.

CO absorption occurs because the background continuum is bright, while the intervening CO gas absorbs photons in rovibrational transitions (rotational + vibrational transitions) without emitting enough radiation to compensate.

CO absorbtion or emission depends on the mass accretion rate, CO band absorption is typically observed in class I or II protostars. FU Orionis-like objects (hot disk midplan with disk atmosphere absorbing), in systems with edge-on disks, and in sources with low or moderate accretion rate.

CO band *emissionn* originates in dense, hot molecular gas, typically located in the inner regions of the circumstellar disks. It traces inner gaseous disk at radii of $\leq 0.1 - 1\text{ AU}$, inside or close to the dust sublimation radius and is occasionally associated with dense disk winds. The temperature and column density required for CO is typically of the order of $\sim 2000 - 5000\text{ K}$ and gas densities $\geq 10^{10}\text{ cm}^{-3}$. CO emission lines are optically thick or moderately optically thick. Emission occurs because CO gas itself becomes an efficient emitter as the higher vibrational levels are populated and the emitted radiation exceeds the background stellar absorption. CO emission band is observed in strongly accreting protostars, high-veiling systems and sources with hot inner disks, and is often detected when the $Br\gamma$ line is observed in emission, It is observed more in class 0 sources than class I.

Here we are talking about the CO overtone band-emissions at $2.3\text{ }\mu\text{m}$, the fundamental CO band $4.6 - 5\text{ }\mu\text{m}$ helps in tracing the warm molecular gas in the inner-envelope or disk winds for embedded class 0 objects, JWST helps with these.

2.0.6 Empirical relations to measure accretion luminosity

Direct modeling of UV continuum excess (primary signature of accretion shocks), is often impossible due to high extinction and continuum veiling, we use empirical relations to measure indirectly the accretion luminosity from line luminosities of different emission lines.

- UV, optical and NIR lines: Hydrogen recombination lines like $Pa\beta$ and $Br\gamma$ and lines from other species like OI, HeI, NaI, CaI, FeI.
- Mid-IR(MIR) lines: JWST has shown that MIR emission lines from species like CO, H_2O , OH, HCN, C_2H_2 and CO_2 as well as high n atomic hydrogen transitions, also correlate with L_{acc}
- UV proxies: In some class 0 protostars, $Ly\alpha$ flux has been linked to the integrated UV continuum to estimate L_{acc} through H_2O photodissociation.

To systematically apply these CTTS-calibrated relations to embedded clas I and FS protostars, we use the following mathematical framework:

1. Bolometric luminosity Assumption: The total L_{bol} and L_{acc} where disk emission is treated as reprocessed accretion energy: $L_{bol} = L_{\star} + L_{acc}$
2. Veiling Parametrization: The unresolved contributions of the disk and envelope are parameterized through the K-band veiling (r_K). This allows the absolute bolometric magnitude (M_{bol}) to be expressed as: The bolometric magnitude is given by:

$$M_{bol} = BC_K + m_K + 2.5 \log(1 + r_K) - A_K - 5 \log \left(\frac{d}{10 \text{ pc}} \right), \quad (3)$$

where BC_K is the bolometric correction in the K -band, m_K is the apparent K -band magnitude, r_K is the K -band excess ratio, A_K is the K -band extinction, and d is the distance.

3. Empirical Line Calibration: The empirical relations calibrated on CTTSs are assumed to be valid for protostars. For the $Br\gamma$ line, this is defined as:

$$\log L_{acc} = a \log L_{Br\gamma} + b, \quad (4)$$

where the calibration constants are $a = 1.19 \pm 0.10$ and $b = 4.02 \pm 0.51$

2.1 Empirical Scaling Relations

Because a substantial fraction of the accretion energy is re-radiated through these broad emission lines, extensive observational campaigns have established tight empirical power-law correlations between line luminosity (L_{line}) and total accretion luminosity (L_{acc}). Notably, works by [Stelzer *et al.* \(2013\)](#) and [Alcalá, J. M. *et al.* \(2017\)](#) leverage X-shooter spectroscopy to parameterize these relationships in the form:

$$\log \left(\frac{L_{acc}}{L_{\odot}} \right) = a \cdot \log \left(\frac{L_{line}}{L_{\odot}} \right) + b \quad (5)$$

where the coefficients a and b are uniquely determined for individual atomic transitions to account for optical depth variations and line formation geometries.

2.2 Extinction and Dereddening Formulations

A significant challenge in YSO spectroscopy is the severe attenuation caused by localized circumstellar environments and interstellar dust. Precise dereddening is essential before line fluxes can be converted to absolute luminosities. Recent advancements, such as the parameter-average extinction laws developed by [?](#) , provide highly accurate, unit-aware formulations for determining specific attenuation variables (A_λ) across a broad wavelength baseline.

2.3 Simulation

We can simulate the cloud/core collapse leading to formation of protoplanetary disk formation by three main methods: Thin disk approximations, Smooth particle Hydrodynamics (SPH) and Adaptive mesh refinement (AMR). I am familiar with using FLASH which is an AMR code that can be used to study magnetohydrodynamics of newtonian fluid structures, hence it is possible to use it for studying protoplanetary disk evolution and formation in varying timescales.

2.3.1 Thin disk (2D models)

Thin-disk approximation assumes that vertical force balance is established on timescales shorter than timescales governing cloud contraction. Isothermal, non-rotating, magnetized cylindrical or spherical clouds quickly reach vertical hydrostatic equilibrium- fast enough to repel the contraction in the direction perpendicular to the field lines caused by ambipolar diffusion. Thus, we can model a thin disc with scale height much smaller than the radial extent, as model cloud, this is one of the simplest ways to numerically simulate the problem of a contracting cloud. It can be used to evolve through entire class I lifetime (of order $\sim Myr$). It does not however take magnetic field geometry into account as well as the vertical disk structure, which is important for evolution in earlier classes, hence, detailed 3D simulations are needed to fully describe the evolution of protoplanetary disks.

2.3.2 Smoothed Particle Hydrodynamics SPH

SPH involves use of lagrangian scheme to solve hydrodynamic equations with properties smoothed over a local kernel, they employ a moving mesh that follows the flow of liquid. This allows to take into account the change in density and morphology of fluid without needing to refine the mesh, therefore, the resolution is automatically best at high particle density regions. Another perk of SPH is the intrinsic conservation properties of motion; linear momentum, angular momentum and entropy are simultaneously exactly conserved independently of configuration, which makes SPH particularly well suited to treat complex 3D geometries. The main limitation of SPH is lack of explicit treatment of physical viscosity, which is addressed by the addition of an artificial viscosity to capture shocks and dissipative processes. Such prescription can be implemented in multiple ways and will effect the angular momentum.

2.3.3 Adaptive Mesh Refinement (AMR)

AMR is an Eulerian scheme which covers the computational domain by a hierarchy of nested grids and are dynamically refined in the regions where higher resolution is required. AMR resolves for both larger and smaller scales; also in contrast with SPH the shocks and discontinuities are easily captured without the need for artificial terms and a Eulerian grid allows for straightforward implementation of non-ideal MHD terms, radiative diffusion, and chemical cooling, with a substantial increase in computational cost limiting the time of such detailed evolution to a max of $10^5 yr$. To make the timestamp prohibitively small near the central object, protostar is replaced by a sink particle, which are unresolved particles that can interact with other particles in the simulation through only gravity. This criteria to transform an area of overly dense gas particles and its neighbors into sink particles, involves not only density but smoothening length, as well as tests to determine if the particles may be in the process of tidal disruption or bouncing, ensuring that sink particles only replace the regions that continue to collapse. This use of sink particles controls computational time and allows to continue the evolution past the first protostars, but has the disadvantage that sink particles introduce regions of unresolvability, where the internal processes such as emission of radiation from the accreting gas and radiation feedback cannot be traced with certainty, and does not allow to separate the inner disk from the protostar itself, which influences the determination of accretion properties.

FLASH-code can implement AMR simulations like the ones shown in paper, it can also do 2D and 3D magnetohydrodynamic simulations but for SPH simulations we need a different code if required, i can check this out as well. **Thin disk** models are optimal to study the long-term evolution of protostars, virtually upto the end of protoplanetary phase (class II), while they don't allow the detailed modeling of accretion and radiative feedback mechanisms; **SPH** is a powerful tool for 3D simulations, with its adaptive resolution and intrinsic conservation properties, but presence of an artificial viscosity limits the resolution of shocks and treatment of radiation and magnetic fields; **AMR** is excellent at resolving the multi-scale nature of collapse and capturing the details of the non-ideal MHD effects, as well as radiation transport and dynamical feedback, but its extreme computational cost reduces the time that the simulation can be done in detail.

2.4 Pure Hydrodynamic simulations

First generation of cloud collapse simulations featured a pure hydrodynamical approach with self-gravity, typically using thin-disk or SPH simulations.

Early papers focused on the onset of instability in the star formation process, focusing on formation of protostar and protoplanetary disk due to the collapse of a single core, become quickly unstable and form spiral, which is formed from continuous infall of envelope. Fragmentation takes place within the spiral arms, forming clumps that occasionally fall into the protostars, causing mass accretion rates as high as $\sim 10^4 M_\odot yr^{-1}$ as opposed to quiescent periods, where accretion rate is of order of $10^{-7} - 10^{-6} M_\odot yr^{-1}$.

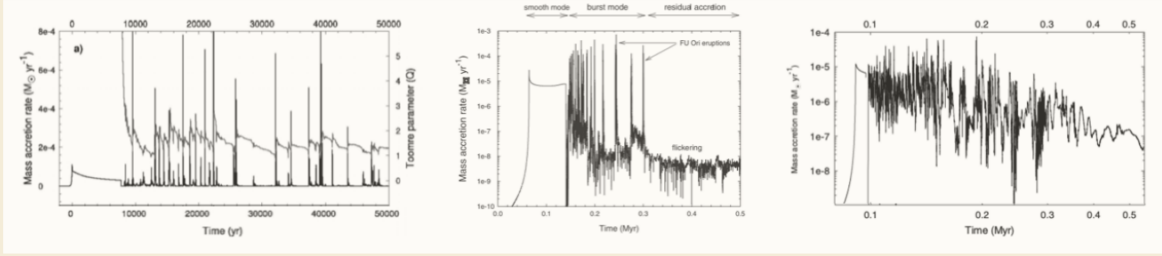


Figure 6. Time evolution of the accretion rate in thin-disc approximation simulations. Adapted from Vorobyov and Basu (2005b, 2006, 2010) (panels from left to right). © AAS. Reproduced with permission.

Figure 2: Mass accretion rate evolution using thin disk approximation simulations. [Fiorellino & Somigliana \(2026\)](#)

2.4.1 Early SPH and grid-based simulations

Early SPH simulations were done in 3D but still revolved around purely hydrodynamical framework coupled to self-gravity and simple barotropic or isothermal equations of states.

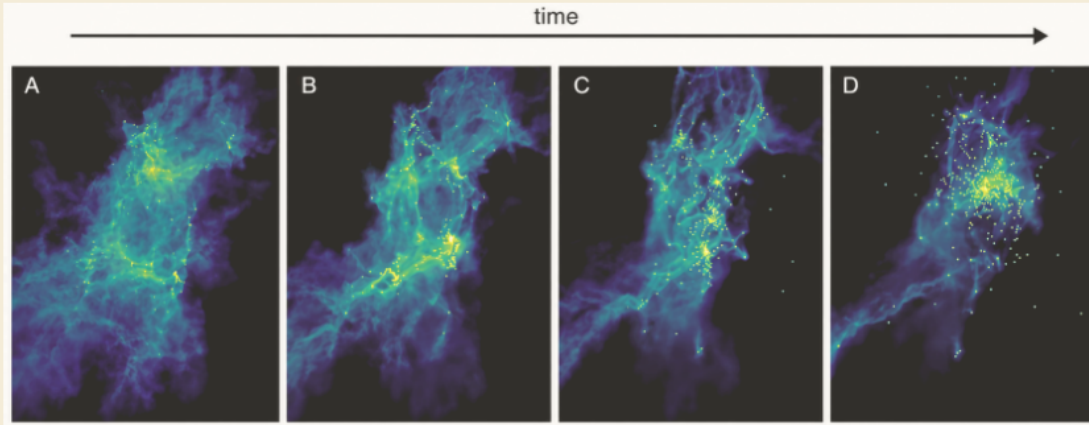


Figure 7. Snapshots of the logarithm of the column density in a $1 \text{ pc} \times 1 \text{ pc}$ region produced from the simulations of Bonnell et al. (2003) at four consequent times from left to right ($1.9, 2.6, 3.4,$ and $4.5 \times 10^5 \text{ yr}$). The color scale spans from a minimum of 0.025 (black) to a maximum of 250 (yellow) g cm^{-3} . Each dot indicates a star..

Figure 3: Snapshots of logarithm of column density in a $1 \text{ pc} \times 1 \text{ pc}$ region produced from simulations at four consequent times from left to right ($1.9, 2.6, 3.4,$ and $4.5 \times 10^5 \text{ yr}$). The colour scale spans from a min of 0.025(black) to a maximum of 250 (yellow) g cm^{-3} . Each dot indicates a star.[Fiorellino & Somigliana \(2026\)](#)

The very first 3D calculation follow collapse of a molecular cloud core to stellar densities, within the framework, protostars formed within clustered environments as opposed to isolated collapse models. These showed that the accretion is a dynamical competitive process, often terminated by ejection rather than smoothly declining; showing that the final mass of the protostar is a multivariable property dependent on it's environment rather than just its

core mass. These showed that non-steady accretion is a generic outcome of hydrodynamic collapse in clustered environments, pinpointing the variability in accretion properties in addition to the instability in the disk itself. The main limitation of the pure hydrodynamic model is that they lack radiation and magnetic fields, therefore, we need better models; radiation hydrodynamics and magnetohydrodynamics to simulate the most accurate model, a capability that FLASH code has.

3 Methodology

3.1 `star-melt`

`star-melt` (Stellar Accretion Mapping With Emission-line Tomography) is a python package designed to extract, identify, and fit emission lines in high-resolution time-resolved stellar spectra. The package automates the analyses of atomic gas kinematics, enabling the tomographic mapping of accretion structures, inner disk winds, and stellar-surface hotspots in YSOs. we will be using `star-melt` for our entire analysis.

We will use `star-melt` to analyse CVSO109 also called Haro5-66, a classical T Tauri star (YSO), located in at a distance of 440 Pc, we use the stellar parameter for this system directly from the paper by [Maucó *et al.* \(2018\)](#), which give us the stellar mass $0.5 M_{\odot}$ a stellar radius of $2.66 R_{\odot}$ and visual extinction of $A_V = 0.01$ magnitudes. we use papers as reference for our own independent analysis.

Observations of this CvsO 109 were carried out using the Ultraviolet and Visual Echelle Spectrograph (UVES) and Xshooter spectrograph on very large telescope (VLT)

- UVES: UVES is a single long-slit spectrograph that divides the spectra into two arms, Red (480 nm) and Blue (450 nm). The observations utilized 0.6-arcsec-wide slits in both arms, achieving a high spectral resolution of roughly $R = 70,000$. This setup provides simultaneous wavelength coverage across 330-450 nm, 480-575 nm, and 585-680 nm.
- Xshooter: XSHOOTER is an 11-arcsec long slit-fed spectrograph providing simultaneous coverage of the wavelength region between 300 nm and 2500 nm. The spectrum is divided into three arms: UVB (300-500 nm), VIS (500-1000 nm), and NIR (1000-2500 nm). Using slit widths of 1.0, 0.4, and 0.4 arcsec for the UVB, VIS, and NIR arms respectively, the instrument yields spectral resolutions of approximately $R = 5400$, 18400, and 11600 in the three arms.

3.1.1 Spectral line extraction and line identification

To automate the extraction and fitting of lines we use `star-melt`, this pipeline itself isolates potential emission lines from continuum subtracted spectra using a sigma detection threshold (it is changeable from 1 sigma all the way to 10 sigma). Once the lines are identified, line profiles are modeled using multi-component Gaussian fit, upto 3 gaussian profiles can be selected including, out of which 2 are positive and one is negative, to capture the redshifted absorption features.

We use multi-epoch UVES data for our analysis, we limit ourself to UVES data for analysis as it has higher resolution than Xshooter, We use a **star-melt** pipeline to extract the raw flux and scale the flux for further analysis. We will use different arms for different analysis and conformation, The flux of different epochs from UVES data for the 326 nm arm is shown below;

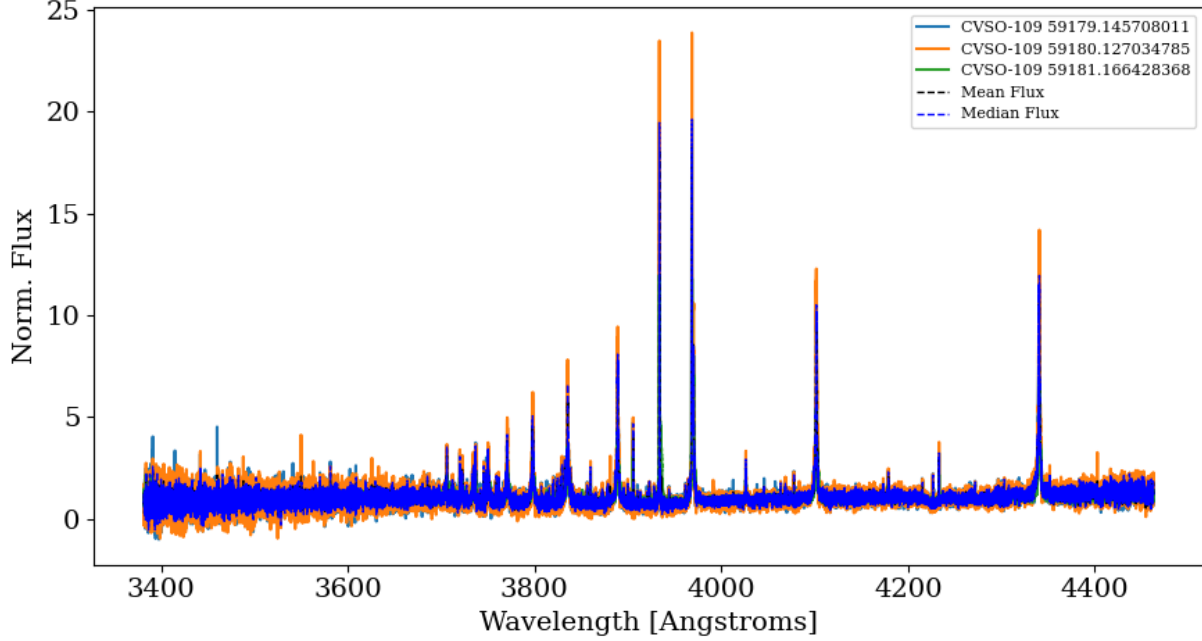


Figure 4: Normalized flux vs wavelength in angstrom produced using **star-melt**, the different colour correspond to different epochs in UVES data, while the *med_flux* gives the median of flux from different exposures.

The emission spectra of the accretion disk, that is the different emission lines from the accretion disk is often veiled by the continuum spectra of the protostar, we need to remove this continuous spectra to get the emission line features in order to find our actual emission lines and fit them using gaussian statistics.

The continuum subtracted emission spectra is found using **star-melt** and is given in the plot below:

After we subtract the continuum spectra, we can finally proceed with identification of lines by matching the peaks observed in the UVES data with the already well established NIST database, Before matching identified spectral line peaks against an atomic reference database (such as NIST), the pipeline corrects the rest wavelength λ_{rest} for the target star's systemic radial velocity (RV)

$$\lambda_{\text{expected}} = \lambda_{\text{rest}} \left(1 + \frac{RV}{c} \right) \quad (6)$$

star-melt is able to identify all the lines by itself, we need to specify the following parameters:

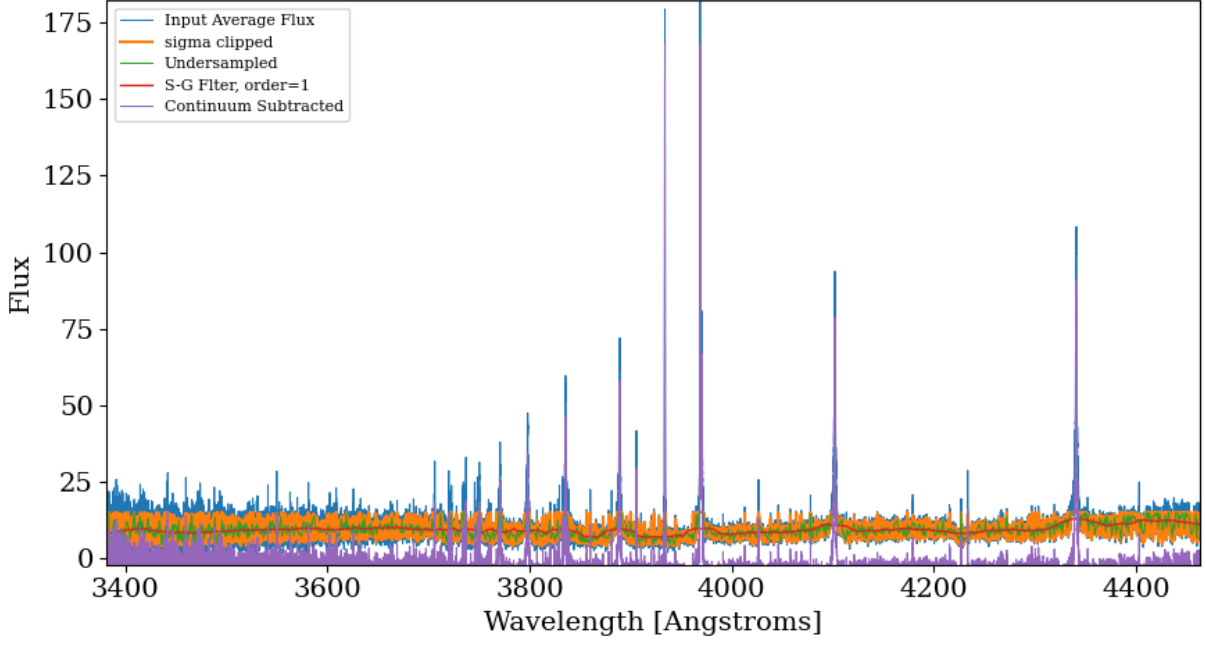


Figure 5: The continuum subtracted emission spectra of CVSO 109 in the arm sensitive to wavelength range 328 to 500 nm (3280 to 5000 Angstrom), normalized to the background spectra.

- **sigma-clipping Threshold slider (σ):** It controls the SNR detection threshold above local continuum level, peaks are flagged as potential emission lines if their amplitude exceeds $N \times \sigma_{cont}$, where σ_{cont} is the standard deviation of the surrounding continuum.
- **wavelength_window: wl_win:**
 - **Local Sampling Domain:** For each wavelength pixel λ_0 , the parameter **wl_win** defines a symmetric localized spectral interval $I(\lambda_0)$:

$$I(\lambda_0) = \left[\lambda_0 - \frac{\text{wl_win}}{2}, \lambda_0 + \frac{\text{wl_win}}{2} \right] \quad (7)$$

- **σ -Clipping:** To prevent real emission lines from inflating the local baseline, positive flux spikes exceeding a min value $I(\lambda_0)$ are temporarily masked out.
- **Continuum Baseline Modeling:** A low-degree polynomial or Savitzky-Golay (SG) filter is fitted across the unmasked continuum pixels in $I(\lambda_0)$ to derive the local continuum flux $F_{cont}(\lambda_0)$.
- **Line Detection Criterion:** The net continuum-subtracted flux $F_{sub}(\lambda_0) = F_{obs}(\lambda_0) - F_{cont}(\lambda_0)$ is evaluated. A candidate emission line is flagged at λ_0 if:

$$F_{sub}(\lambda_0) \geq N \times \sigma_{cont} \quad (8)$$

where σ_{cont} represents the standard deviation of the local continuum noise.

- **Impact of Parameter Adjustments (wl_win Variation):**

- **Under-sized Window ($wl_win < \Delta\lambda_{line}$, e.g., 0.1–0.3 Å):**

- * *Logic:* The sampling span is narrower than the full width of broad emission profiles ($\Delta\lambda_{line}$).
- * *Effect:* The fitted baseline curves up into the broad emission profile, interpreting the line wings as continuum. This artificially reduces F_{sub} below $N \times \sigma_{cont}$, causing the pipeline to miss broad permitted lines (e.g., H α , H δ , Ca II K).

- **Optimal Window ($wl_win \approx few \times \Delta\lambda_{line}$, e.g., 0.5–1.5 Å):**

- * *Logic:* The window balances local baseline flexibility with profile coverage.
- * *Effect:* The estimated baseline covers individual lines while tracking the local continuum slopes, yielding accurate peak detections.

- **Over-sized Window ($wl_win \gg \Delta\lambda_{line}$, e.g., 3.0–5.0 + Å):**

- * *Logic:* The sampling domain spans across multiple neighboring transitions, echelle order boundaries, or atmospheric absorption features.
- * *Effect:* The baseline becomes very rigid, it's unable to adapt to local continuum depressions (such as broad photospheric absorption features), producing incorrect positive detection peaks.

- **a_tol:** It defines the maximum allowable wavelength shift (in angstrom Å), between observed emission peak and a candidate atomic transition in the reference database, matching only those transitions where $|\lambda_{obs} - \lambda_{expected}| \leq a_tolerance$

3.1.2 Detected lines in the continuum subtracted spectra

The lines that are confirmed in our continuum subtracted spectra, when taking a $\sigma = 5$ and $wl_min = 0.9$ with $a_tol = 4$ is shown below;

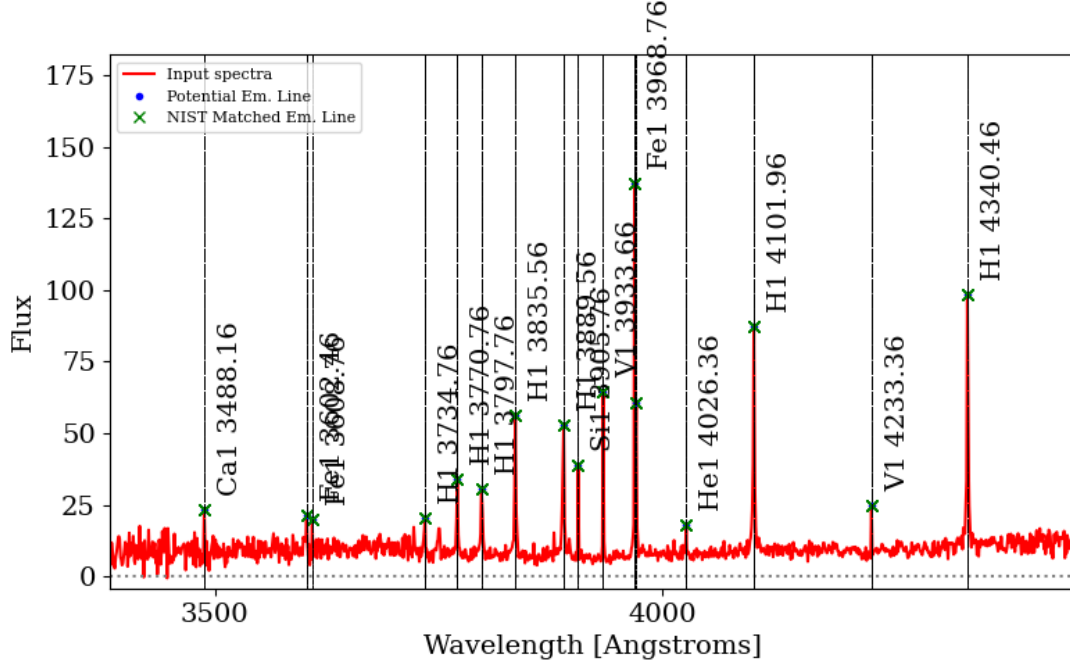


Figure 6: All the identified emission lines in CVSO-109 from UVES spectra, for the 328 nm arm, matched with NIST database, for the chosen values $\sigma = 5$ and $wl_min = 0.9$ with $a_tol = 4$

3.2 Gaussian fitting

The lines identified from the line emission spectra have multiple matches with the NIST database, in order to choose the actual data in our spectra, we use the following norms: we choose the relative wavelength, rv_wl that has the highest aki number; aki number is the probability of a transition happening, the higher the aki number the greater the chance of that emission happening and we choose those emission lines which have the smallest velocity difference, vel_diff .

we use the (Campbell-White *et al.*, 2021) paper to choose our lines for our initial analysis, the goal is to use the lines that have a good resolution, and using the emperical relation

$$\log \left(\frac{L_{acc}}{L_{\odot}} \right) = a \cdot \log \left(\frac{L_{line}}{L_{\odot}} \right) + b \quad (9)$$

Alcalá, J. M. *et al.* (2017) we find the mass accretion rates from the chosen emission lines.

For our initial analysis we use the following lines;

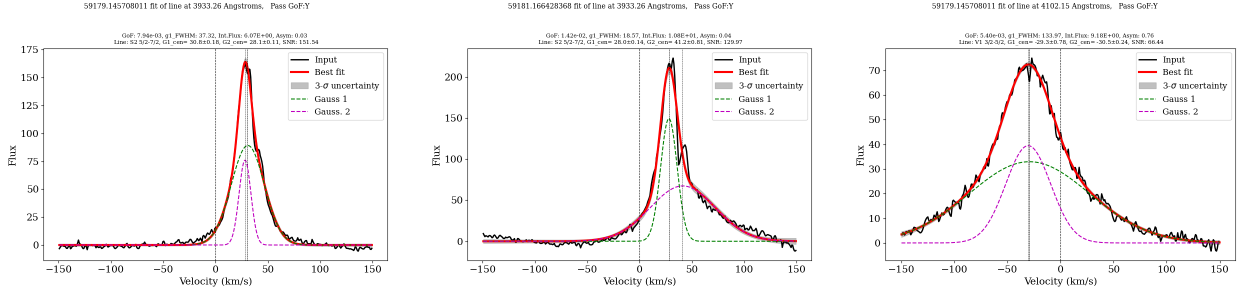
- 3933.66 Å, which had the relative wavelength, $rv_wl=3933.48$ Å and $obs_wl_air = 3933.26$ Å
- 4101.96 Å, which had the relative wavelength, $rv_wl=4102.38$ Å and $obs_wl_air = 4102.15$ Å

While `star_melt` accelerates initial line sorting, numerical artifacts can emerge during the curve-fitting loop. For instance, severe floating-point scaling adjustments implemented

internally to prevent underflow errors often render the raw output integrated flux column (`int_flux`) unitless. To establish dimensional accuracy, our modified pipeline bypasses automated scaling by combining the scale-invariant Equivalent Width with a localized median continuum measurement:

$$F_{\text{line}} = |\text{EW}| \times F_{\text{continuum}} \quad (10)$$

The fitted gaussian plot for the above two lines are shown in the plot below:



(a) Gaussian fit for the line 3933.26 Å for the mjd 59179.145708011 we fit two Gaussians to the emission line profile, by changing the slider, **star-melt** automatically fits the best gaussian profile using a goodness of fit (gof) parameter, we chose this to gof be 0.8

(b) Gaussian fit for the line 3933.26 Å for the mjd 59181.166428368, we fit two Gaussians to the emission line profile by changing the slider, **star-melt** automatically fits the best gaussian profile using a goodness of fit (gof) parameter, we chose this gof to be 0.8

(c) Gaussian fit for the line 4102.15 Å for the mjd 59179.145708011, we fit two Gaussians to the emission line profile by changing the slider, **star-melt** automatically fits the best gaussian profile using a goodness of fit (gof) parameter, we chose this gof to be 0.8

3.2.1 Fitting H α (6563.02) and He I (5876.61) lines

H α and He I lines are very prominent emission lines observed in the UVES 576 nm arm, we choose a $\sigma = 6$ and `wl_win = 1.20`, to exactly filter out these lines and perform the same analysis as done above on these, that is we fit gaussian curves on these lines using **star-melt** and derive the mass accretion rate. For fitting we feed the `obs_wl_air` with `radvel = 16.9048`, taken directly from the [Maucó et al. \(2018\)](#) paper. We fit 2 gaussian curves on the `df_av_norm`, which is the normalized flux to , background continuum, with a gof parameter = 0.80. Using the aki number and velocity difference, we fit `obs_wl_air = 6561.93 Å` and `5875.62 Å` respectively.

The continuum subtracted flux for the 576 UVES arm is given below:

To identify the lines in our continuum subtracted flux, we choose a σ of 6 and a `wl_win = 1.2` so isolate only the huge peaks in the spectra, the plot for this is shown below:

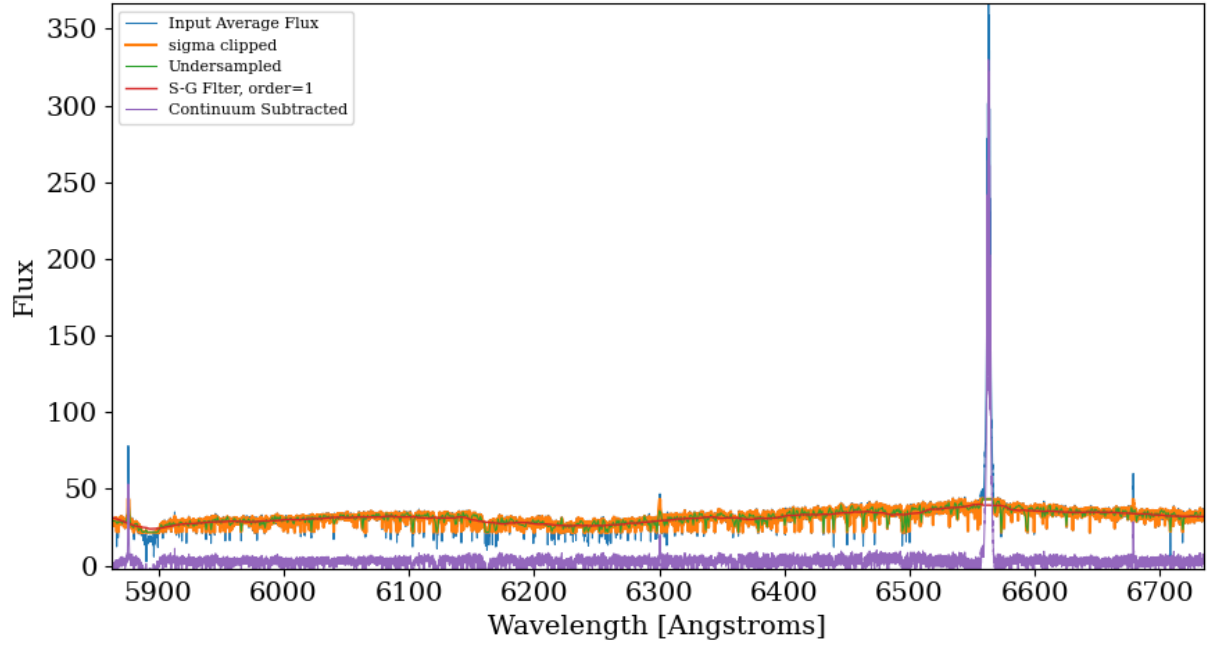


Figure 8: The continuum subtracted flux for NIR UVES arm, we can clearly see a huge H α peak in this continuum subtracted plot.

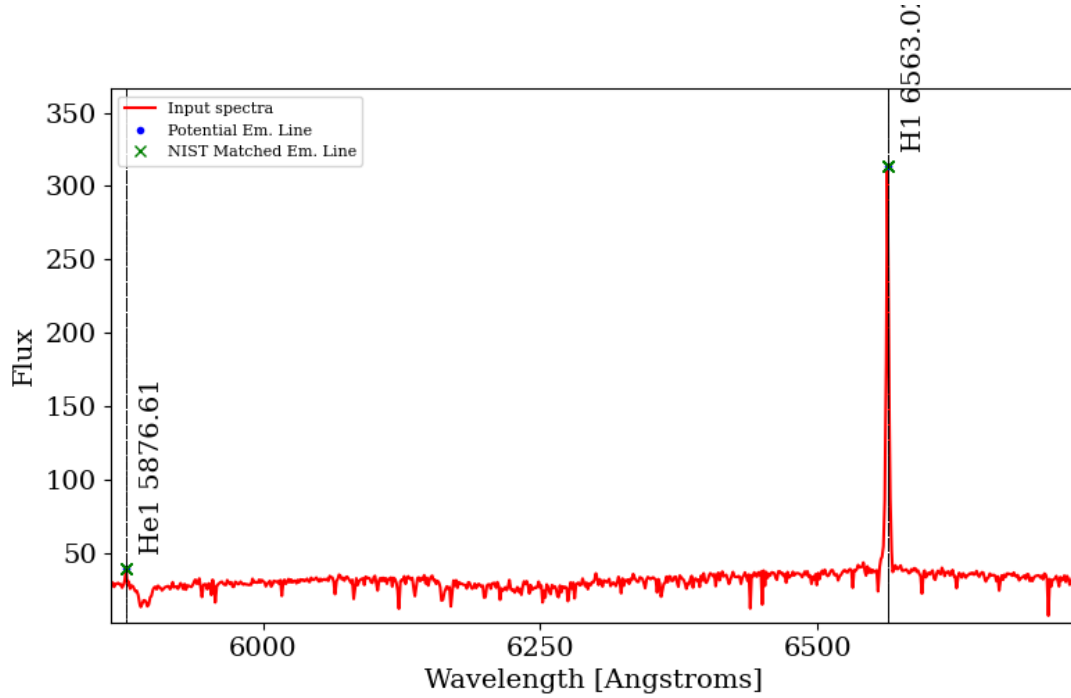
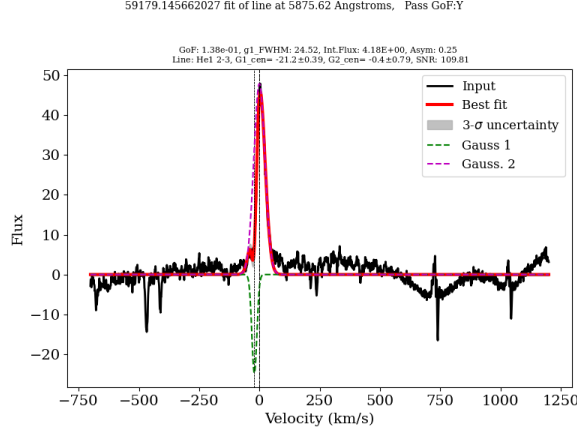
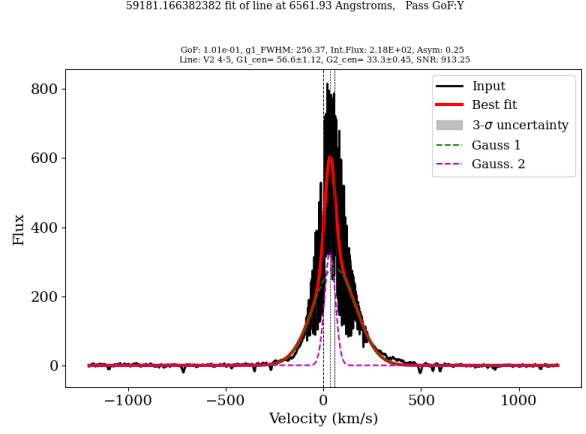


Figure 9: Detected lines in the continuum subtracted flux in NIR UVES arm, we choose a $\sigma = 6$ and $wl_win = 1.2$, clearly identifying the H α and the He I emission peaks

We use similar method to fit the gaussian curves on these lines as described above. The fitted gaussian curves for the lines are shown below, we show just 1 epoch data for each line, the accretion rate due to different epochs for each line will be shown as a table in the discussion section of the report,



(a) Gaussian fits for the line 5875.62 Å we have fit 2 gaussian curves on the line profile with a gof of 0.6



(b) Gaussian fit for the line 6561.93 Å, we fit two gaussian curves on this line with a gof parameter of 0.6

4 Result and Conclusion

The calculated values of mass accretion rate from line luminosity after doing the gaussian fitting using `star-melt` for the fitted lines is given in the table below:

Table 1: Extracted Line Parameters, Luminosities, and Mass Accretion Rates

w_0 (Å)	MJD	λ_{RV} (Å)	Δv (km s ⁻¹)	v_{g1} (km s ⁻¹)	v_{g2} (km s ⁻¹)	EW (Å)	F_{dered} (erg s ⁻¹ cm ⁻²)	Coefficients (a, b)	L_{line} ($L_{\odot}$)	$\dot{M}_{acc}$ ($M_{\odot}$ yr ⁻¹)
3933.66	59181.17	3934.09	32.89	-15.20	—	-21.43	1.45×10^{-15}	(1.05, 2.30)	3.36×10^{-5}	2.07×10^{-10}
4101.96	59179.15	4102.38	30.69	-29.31	-30.45	-5.36	9.29×10^{-15}	(1.06, 2.92)	5.60×10^{-5}	5.51×10^{-9}
4101.96	59180.13	4102.38	30.69	-12.97	-132.90	-9.81	1.31×10^{-14}	(1.06, 2.92)	7.90×10^{-5}	7.95×10^{-9}
He I ($\lambda_0 = 5876.61$ Å)										
5876.61	59179.15	5875.95	33.67	-21.16	-0.39	-1.93	4.22×10^{-15}	(1.13, 3.51)	2.60×10^{-5}	4.42×10^{-9}
5876.61	59180.13	5875.95	33.67	21.16	6.68	-6.83	1.44×10^{-14}	(1.13, 3.51)	8.70×10^{-5}	1.77×10^{-8}
Hα ($\lambda_0 = 6563.02$ Å)										
6563.02	59181.17	6562.30	32.89	56.65	33.26	-37.33	2.19×10^{-13}	(1.12, 1.50)	1.33×10^{-3}	4.01×10^{-9}

These values agree, when compared with the [Maucó *et al.* \(2018\)](#) paper, where their found value of $\dot{M}_{acc}$ was $\sim 6.71 \times 10^{-9} M_{\odot} \text{yr}^{-1}$

5 Future scope of work

We have yet to implement the detailed profile analysis of broadband component, as our particular choice of lines will determine what accretion rate we will be getting, we can take

the redshifted component in a broadband emission line to get the mass accretion rate and compare that with the mass accretion rate from direct tracers like $H\alpha$. Future work will involve using 3D simulations using codes like Rad-mc 3D and FLASH in order to simulate the accretion disk evolution and match the parameters that are observationally measured, we can compare the and match the simulation data with observational data and fit the models that correctly match the observations.

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